

CLOUD_PRACTICAL:

TASK 01:

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&i...
AWS
Q Search [Alt+S]
United States (N. Virginia) Ayush Chinchore (059898286514) Ayush%20Chinchore

ubuntu@ip-172-31-27-75:~$ sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan

Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse rootlesskit zfs-fuse | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 12
  Download size: 74.0 MB
  Space needed: 281 MB / 4681 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 containerd amd64 2.2.2-0ubuntu1 [28.0 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dns-root-data all 2025080400build1 [6022 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 dnsmasq-base amd64 2.92-1ubuntu0.3 [441 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 docker.io amd64 29.1.3-0ubuntu4.1 [35.5 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ubuntu-fan all 0.12.17 [34.3 kB]
Fetched 74.0 MB in 1s (76.6 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1build1_amd64v3.deb ...
Unpacking pigz (2.8-1build1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-4ubuntu3_amd64v3.deb ...
Unpacking bridge-utils (1.7.1-4ubuntu3) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.4.0-0ubuntu1_amd64v3.deb ...
Unpacking runc (1.4.0-0ubuntu1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_2.2.2-0ubuntu1_amd64v3.deb ...
```

i-05e32ebb7513db95c (Cloud-Practical)

PublicIPs: 54.208.102.190 PrivateIPs: 172.31.27.75

```
ubuntu@ip-172-31-27-75:~$ sudo systemctl enable docker
ubuntu@ip-172-31-27-75:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-13 05:18:08 UTC; 1min 46s ago
   Invocation: 9e168c98f3214196b0ff512f462f5f01
   TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 2793 (dockerd)
   Tasks: 9
   Memory: 26.6M (peak: 27.1M)
   CPU: 387ms
   CGroup: /system.slice/docker.service
           └─2793 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.210864040Z" level=info msg="Restoring containers: start."
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.256669320Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.262880089Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.615403016Z" level=info msg="Loading containers: done."
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.623536586Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 containerd-snapshotter-tru
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.623765500Z" level=info msg="Initializing buildkit"
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.668469935Z" level=info msg="Completed buildkit initialization"
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.67394942Z" level=info msg="Daemon has completed initialization"
Jun 13 05:18:08 ip-172-31-27-75 systemd[1]: Started docker.service - Docker Application Container Engine.
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.674819191Z" level=info msg="API listen on /run/docker.sock"
lines 1-23/23 (END)
```

i-05e32ebb7513db95c (Cloud-Practical)

PublicIPs: 54.208.102.190 PrivateIPs: 172.31.27.75

```
Jun 13 05:18:08 ip-172-31-27-75 dockerd[2793]: time="2026-06-13T05:18:08.674819191Z" level=info msg="API listen on /run/docker.sock"
ubuntu@ip-172-31-27-75:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-27-75:~$
```

```
ubuntu@ip-172-31-27-75:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
6376488be516: Pull complete
72c03230f136: Pull complete
d4dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-27-75:~$
```

i-05e32ebb7513db95c (Cloud-Practical)

PublicIPs: 54.208.102.190 PrivateIPs: 172.31.27.75

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Last login: Sat Jun 13 05:16:04 2026 from 18.206.107.29
ubuntu@ip-172-31-27-75:~$ sudo docker run -d --name nginx-container -p 80:80 nginx
f6f8f80dc89b90b2f8037a4fdbff89d2e276cec23558d9f97b15a19eeda31b7
ubuntu@ip-172-31-27-75:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
f6f8f80dc89b   nginx    "/docker-entrypoint...." 12 seconds ago Up 11 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp nginx-container
ubuntu@ip-172-31-27-75:~$ curl ifconfig.me
54.208.102.190ubuntu@ip-172-31-27-75:~$

```

TASK 02:

```

← → ↻ us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-
aws [Search] [Alt+S] Ask Amazon Q
ubuntu@ip-172-31-27-75:~$ sudo docker volume create studentdata
studentdata
ubuntu@ip-172-31-27-75:~$ sudo docker volume ls
DRIVER      VOLUME NAME
local       studentdata
ubuntu@ip-172-31-27-75:~$
local       studentdata
ubuntu@ip-172-31-27-75:~$ sudo docker run -it --name student-container -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@03300d4b909f3:/#

```

TASK 03:

```

aws [Search] [Alt+S]
ubuntu@ip-172-31-27-75:~$ mkdir practicealexam
ubuntu@ip-172-31-27-75:~$ cd practicealexam
ubuntu@ip-172-31-27-75:~/practicealexam$ nano Dockerfile
ubuntu@ip-172-31-27-75:~/practicealexam$ sudo docker build -t practicealexam
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

docker: 'docker build' requires 1 argument
Usage:  docker build [OPTIONS] PATH | URL | -

Run 'docker build --help' for more information
ubuntu@ip-172-31-27-75:~/practicealexam$
ubuntu@ip-172-31-27-75:~/practicealexam$ sudo docker images

IMAGE                ID                        DISK USAGE   CONTENT SIZE  EXTRA
nginx:latest         608a100c7165             241MB        66MB          U
ubuntu:latest        f3d28607ddd7             160MB        45.3MB        U
ubuntu@ip-172-31-27-75:~/practicealexam$

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